

Home Examination

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1 Introduction

The studied system consists of a simple gas fed turbine fed by CH_4 fuel, shown in Figure 1.

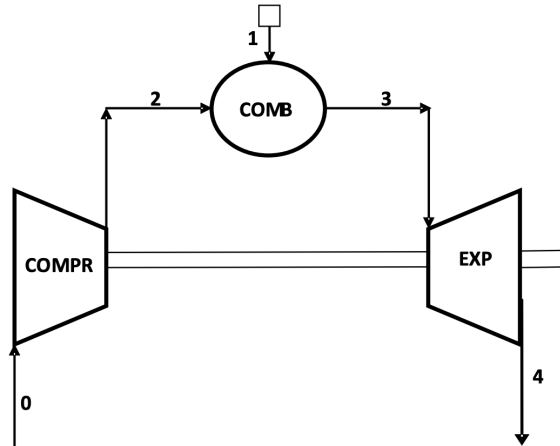


Figure 1: Physical model of the power plant

This system consists of a compressor, a combustor mixing air and fuel CH_4 , and a turbine, expanding the resultant gas mixture and producing useful work. The goal of the study is to evaluate, through Matlab, a thermoeconomic optimization. The procedure is done setting the energy product of the system, which is provided in the exam sheet and correspond to $W_{el} = 50 \text{ MW}$.

The goal is to find the optimum values of β (compressor pressure ratio), η_c (compressor isentropic efficiency), η_t (turbine isentropic efficiency) and of T_3 (turbine inlet temperature), which minimize the objective function that accounts for the annual cost of the energy system. All these parameters are "stored" in the X decision variable. The objective function is represented by the following equation:

$$\min_x F(x) = \min_x \left\{ \tau * N \left[\sum_i Z_i(x) + C_f(x) \right] \right\}. \quad (1)$$

where,

- $Z_i(x)$ are the recovery costs of the components on an annual basis, taking into account the investment costs ($C_i(x)$), maintenance factor (ϕ), number of operation hours per year (N), capital recovery factor (CRF) and τ assumed of 3600 s/h .
- $C_f(x)$ is the cost of fuel

The investment costs per component $C_i(x)$ functions are provided in the exam's guidelines, together with the economic parameters used in the $C_i(x)$ equations. Other costs will be added to the equation during the study, such as the fixed costs, costs of the stack and costs of the emissions.

2 Thermo-economic optimization of the plant in nominal conditions

The analysis in nominal conditions is performed through two MatLab scripts, called *exam25.m* and *optimize_GT.m*. The two MatLab scripts work together to find the minimum cost design of the plant. The first script, *exam25.m*, takes as input the set of five decision variables (compressor pressure ratio, compressor and turbine isentropic efficiencies, preheater temperature, and turbine inlet temperature) and uses basic cycle relations to compute mass flows, work outputs, and the five components' costs terms (compressor, combustor, turbine, exhaust stack and

fuel). It then combines these into a single annualized cost objective function while utilizing an inequality constraint ($T_2 < T_3$). The second script, *optimize_GT.m*, provides an initial guess and bounds for those variables, calls MatLab's function *fmincon* to minimize the cost returned by *exam25.m* under the specified constraint, and finally reports the optimal design variables and minimum cost (with the option to display each individual cost contribution). To perform the analysis, an initial guess of the X values was made, particularly:

- Beta (pressure ratio): 15
- Compressor efficiency: 0.8468
- Turbine efficiency: 0.8786
- Preheat temp T2 [K]: 914.28
- Turbine inlet temp T3 [K]: 1492.63

Secondly, some variable bounds needed to be set.

- Beta (pressure ratio): $10 < \beta < 20$
- Compressor efficiency: $0.5 < \eta_c < 0.89$
- Turbine efficiency: $0.5 < \eta_t < 0.91$
- Preheat temp T2 [K]: $800 < T_2 [K] < 1200$
- Turbine inlet temp T3 [K]: $1300 < T_3 [K] < 1550$

The optimal values of the parameters that have been found after the simulation are the following:

- Beta (pressure ratio): 19.94
- Compressor efficiency: 0.85
- Turbine efficiency: 0.90
- Preheat temp T2 [K]: 1009.33
- Turbine inlet temp T3 [K]: 1468.69

With a final minimum annual costs value of 29,000,079.14 *USD/year*.

The results show average values for a 50 MW system, demonstrating consistency. For example, Siemens GT-800 show a pressure ratio of 19.8 [1].

Moreover, to get a better overview, the costs of the single components were "printed", and are (in \$:)

- C1 (compressor): 5,642,877.886
- C2 (combustor): 137,508.765
- C3 (turbine): 6,296,846.550
- C_{stack} (exhaust): 207,995.746
- C_{fuel} : 26,534,838.766

It can be seen that while the combustor represents the less impactful component, the cost of fuel used in the its relative process it is much more higher. The results show how there could be an important margin of improvement of the system, from the the economic point of view, in the combustor.

3 Sensitivity analysis varying the cost of fuel (CH_4)

A sensitivity analysis varying the cost of fuel is addressed using two new scripts: *exam25_with_fuel_cost.m* and *fuel_cost_sensitivity.m*. *exam25_with_fuel_cost.m* has the same structure as *exam25.m* with the only differences that at the beginning, the function accepts *cf* (cost of fuel) as a second input, being no more hard-coded; on the other side, *exam25_with_fuel_cost.m* builds a vector of fuel-cost multipliers from -100% to $+100\%$ of the base value, to then loop over these multipliers calling *fmincon* each time and storing the resulting *x* in a matrix. After the loop, it creates five separate plots of each decision variable versus the fuel-cost multiplier, which are here shown.

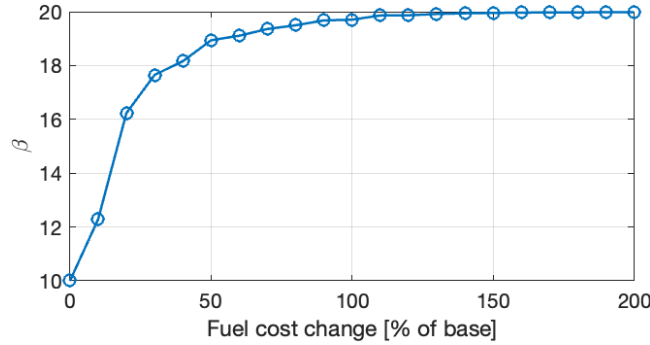


Figure 2: β vs. fuel costing

As the fuel cost increases from -100% to $+100\%$ of its base value, the optimal compressor pressure ratio rises fastly at first and then stabilizes near its imposed upper bound. When fuel is very cheap (at or below zero cost), the optimizer selects a low β (around 10–12) to minimize capital expenditure on the compressor and turbine. As burning fuel becomes more expensive, however, increasing β yields higher cycle thermal efficiency and reduces fuel burn, so the cost-minimizing design shifts toward a higher pressure ratio (up to 20). Closely to a 50% increase in fuel cost, the benefit of further boosting β diminishes (and the bound of 15–20 is reached), so the curve become flat.

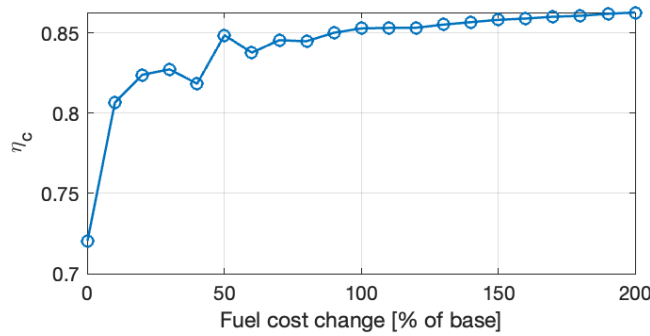


Figure 3: η_c vs. fuel costing

The optimal compressor efficiency rises with fuel price. At very low fuel costs, a relatively inefficient (and cheaper) compressor (0.72–0.80) is preferred, due to non important fuel savings. As the fuel cost increases, investing in a higher-performance compressor becomes worthwhile: η_c quickly jumps to 0.83–0.85, then gradually approaches the upper bound (0.86–0.87) as fuel cost doubles. Infact, it is better to spend on a performative hardware only when it meaningfully reduces the ongoing fuel expense.

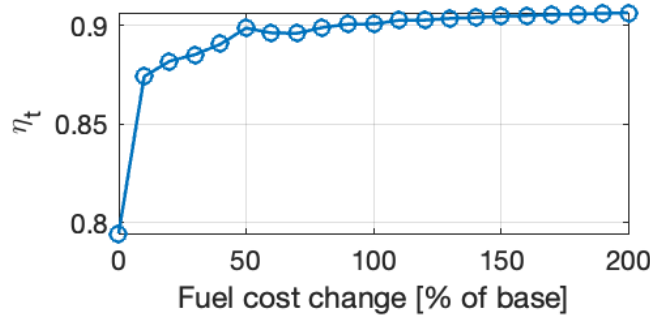


Figure 4: η_t vs. fuel costing

Similar to the compressor, the turbine's optimal efficiency increases with fuel cost and moving from roughly 0.79 at the cheapest fuel to about 0.89 at the most expensive. Initially, fuel-cost sensitivity is very high. After about a 50% fuel-cost increase, it approaches its upper bound (0.90–0.91), meaning that further efficiency gains no longer justify their extra capital cost.

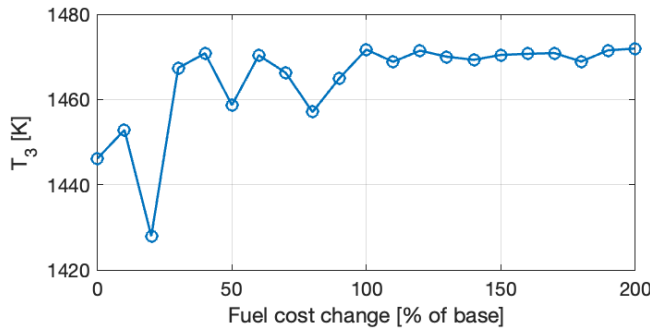


Figure 5: T_3 vs. fuel costing

The turbine inlet temperature shows a general upward trend, rising from about 1445 K when fuel is cheaper to 1470 K at higher fuel prices, which boosts cycle efficiency via a wider temperature difference. However, some small local dips and peaks can be seen, showing the non linear relation between η_t and the fuel's price. Overall, the design pushes T_3 higher when fuel is expensive, then levels off as it approach the upper bound, consistent with the diminishing returns of higher combustion temperatures.

4 Optimal results considering CO and NO_x emissions penalty factors

From literature, values of 51 g/GJ for NO_x and 26 g/GJ for CO have been found [2].

To convert the emission factors into annual penalty costs, the following procedure was assessed. First, the given emission rates were converted from grams per gigajoule to kilograms per gigajoule:

$$51 \frac{\text{g}}{\text{GJ}} = 0.051 \frac{\text{kg}}{\text{GJ}}, \quad 26 \frac{\text{g}}{\text{GJ}} = 0.026 \frac{\text{kg}}{\text{GJ}}.$$

Next, the fuel-energy flow rate were computed in GJ/s by multiplying the fuel mass flow G_{fuel} [kg/s] by the lower heating value LHV [kJ/kg] and dividing by 10^6 :

$$\dot{E}_{\text{fuel}} = \frac{G_{\text{fuel}} \times LHV}{10^6} \quad [\text{GJ/s}].$$

Subsequently, $\dot{E}_{\text{fuel}}$ was multiplied by the annual operating time N [h/yr] converted to seconds ($N \times 3600$), thus obtaining the total fuel energy consumed per year in GJ. Multiplying this by the emission factors (kg/GJ) yields the annual pollutant masses:

$$m_{\text{NO}_x} = 0.051 \times \dot{E}_{\text{fuel}} \times (N \cdot 3600), \quad m_{\text{CO}} = 0.026 \times \dot{E}_{\text{fuel}} \times (N \cdot 3600).$$

Finally, the pollution-penalty factors were applied $fp_{\text{NO}_x} = 1.1$, $fp_{\text{CO}} = 7.0$ and unit costs $c_{\text{NO}_x} = 5$ \$/kg, $c_{\text{CO}} = 3$ \$/kg to obtain the annual dollar-per-year penalties:

$$C_{\text{NO}_x} = fp_{\text{NO}_x} c_{\text{NO}_x} m_{\text{NO}_x}, \quad C_{\text{CO}} = fp_{\text{CO}} c_{\text{CO}} m_{\text{CO}}.$$

These values are then added directly into the total objective function, which result in a new objective function of 31,623,530.59USD/year. Nevertheless, the new optimal design variables now are:

- Beta (pressure ratio): 19.93
- Compressor efficiency: 0.85
- Turbine efficiency: 0.89
- Preheat temp T2 [K]: 1183.20
- Turbine inlet temp T3 [K]: 1470.19

5 Sensitivity analysis varying CO and NO_x environomic costs

A sensitivity analysis varying the cost of the emissions is addressed using two new scripts: *exam25_emiss_sensitivity* and *emissions_sensitivity*. The procedure is the same of section 3, including some changes in the code.

Firstly, above are shown the impacts of NO_x 's price changes.

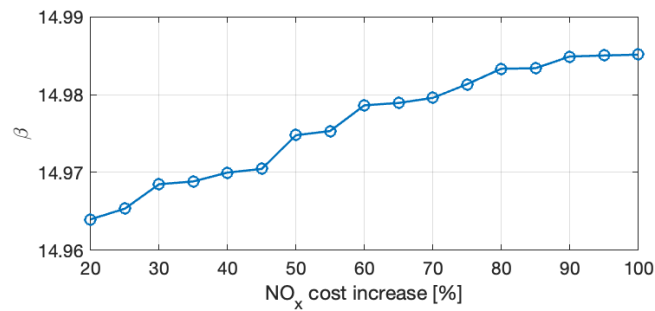


Figure 6: β vs. NO_x 's price

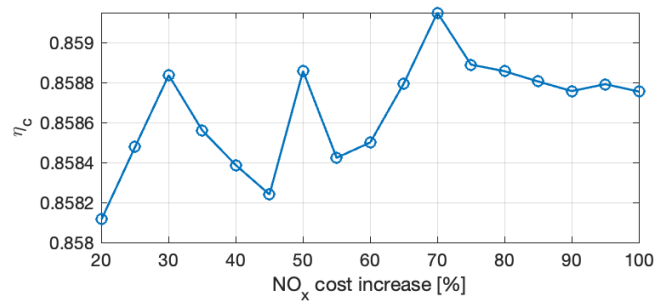


Figure 7: η_c vs. NO_x 's price

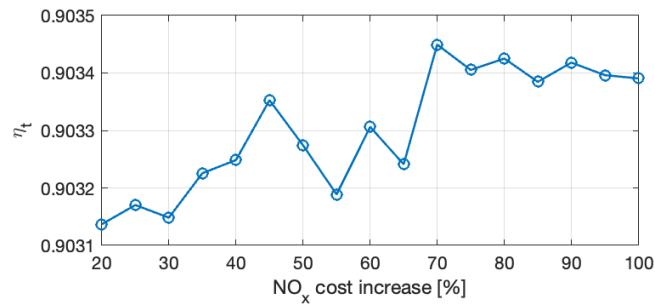


Figure 8: η_t vs. NO_x 's price

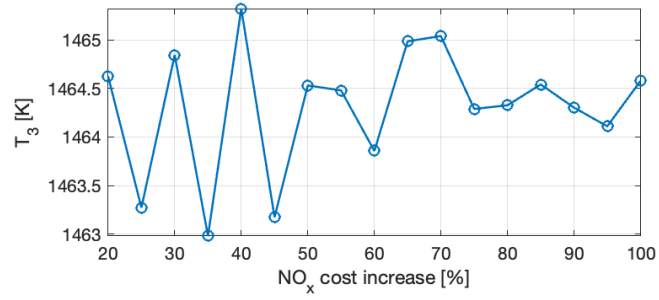


Figure 9: T_3 vs. NO_x 's price

Secondly, above are shown the impacts of CO 's price changes.

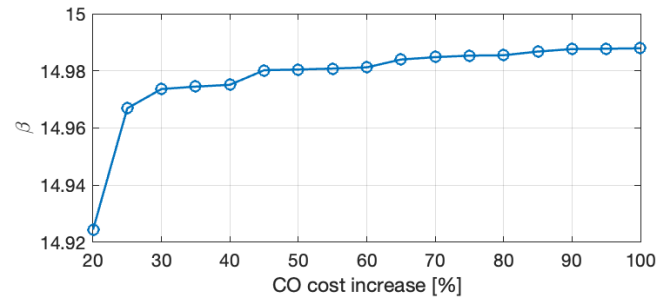


Figure 10: β vs. CO 's price

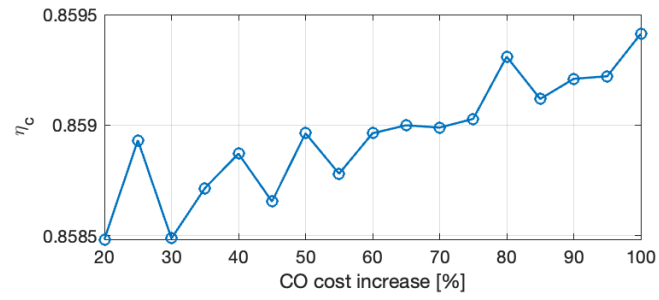


Figure 11: η_c vs. CO 's price

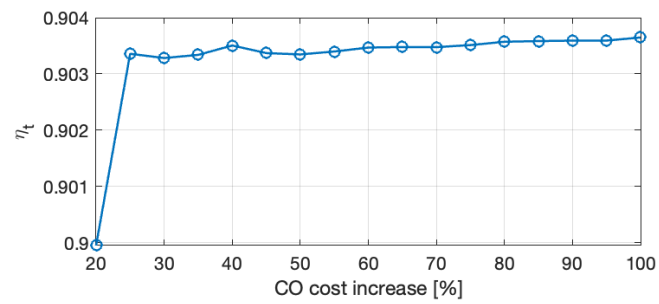


Figure 12: η_t vs. CO 's price

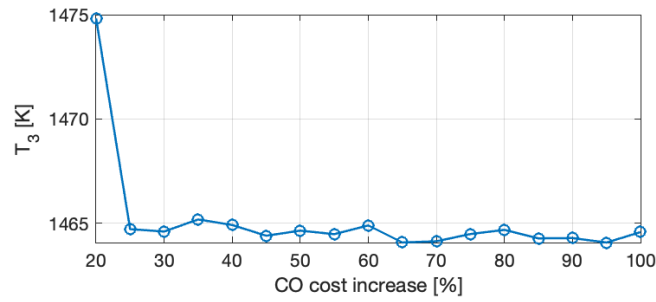


Figure 13: T_3 vs. CO 's price

In both cases, only minor impacts result (Y values), showing that changing the emissions' prices is not relevant to the design of the system. The only variable that shows a trend is the inlet temperature of the turbine, in case of varying the CO 's emission costs. In fact, it is immediately shown a drop in its value from 1475 K to 1465 K, which then stabilizes. T_3 , being the temperature after the combustion process, is strictly related with the emissions.

6 Comparison

Comparing the two policies, thus raising the price of CH_4 versus imposing a tax per kilogram on CO and NO_x emissions, show that their impact is completely different. While doubling the cost of fuel drives large, smooth increase in pressure ratio (from roughly 10 to 20), compressor and turbine efficiencies (by several percentage points), and turbine inlet temperature, this does not happen when changing the price of emissions. Subsequently, the rise in cost of fuel reflects a strong economic incentive to invest in more performative components to cut fuel's costs. On the other hand, rising the emission's costs seems to not be impactful, varying the design parameters of small percentages. To have an impact, these taxes should match unrealistically values, telling us that a tax on fuel is far more effective.

References

- [1] Siemens Energy. *SGT-800 gas turbine*. Tech. rep.
- [2] EMEP/EEA. *Air Pollutant Emission Inventory Guidebook*. Tech. rep. 2023.